

# SERAT M. SAAD

Ph.D. Student in Astronomy, The Ohio State University

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## RESEARCH INTERESTS

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**Dynamics:** binary stars, radial migration, Modified Newtonian Dynamics (MOND)

**Stellar Spectroscopy:** chemical abundances, metal-rich stars, star spots, young stellar objects

**AI for Astronomy:** LLM agents, agent-callable tools, retrieval-augmented generation (RAG)

## EDUCATION

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Ph.D. in Astrophysics (in progress), The Ohio State University 2025-Present  
*Advisors: Prof. Yuan-Sen Ting and Prof. Krzysztof (Kris) Stanek*

A.B. in Physics & Applied Mathematics, Vanderbilt University 2021-2025  
*Thesis: Understanding Accretion Activity in Young Stars using SDSS-V BOSS Spectra*  
*Advisor: Prof. Keivan Stassun*

## HONORS & AWARDS

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University Distinguished Fellowship, The Ohio State University 2028-2029  
*graduate-school fellowship supporting doctoral study*

University Fellowship, The Ohio State University 2025-2027  
*graduate-school fellowship supporting doctoral study*

Hult Prize, National Finalist 2026  
*for a student venture expanding access to personalized education*

Larry R. Cathey Award, Vanderbilt University 2025  
*annual departmental award for the top graduating student studying astronomy*

Highest Honors in Astronomy, Vanderbilt University 2025  
*for distinction in the undergraduate honors thesis and its defense*

International Full-ride Student Grant, Vanderbilt University 2021-2025  
*full four-year scholarship covering all costs of undergraduate study*

Dean's List, Vanderbilt University 2021-2025  
*eight semesters of academic distinction*

Pi Mu Epsilon, Vanderbilt University 2024  
*member of the national mathematics honor society*

Sigma Pi Sigma, Vanderbilt University 2023  
*member of the physics honor society for academic achievement in physics*

Buchanan Fellowship, Vanderbilt University 2022  
*supported a data-science project applying NLP to British periodical collections*

Silver Medal, International Olympiad on Astronomy and Astrophysics 2020  
*represented Bangladesh at IOAA 2019 and the Global e-Competition (GeCAA) 2020*

## PUBLICATION SUMMARY

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12 papers, 7 as first author; 5 published in refereed journals, 1 in press, and 6 currently under review.  
102 citations from 99 citing papers ([NASA ADS Library](#)).

## FIRST AUTHOR PUBLICATIONS

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7. **Serat M. Saad** & Yuan-Sen Ting (2026)  
“More than Half of Recent Astronomy Papers are Written with Language-Model Assistance”  
*Nature Astronomy, in press*
6. **Serat M. Saad** & Yuan-Sen Ting (2026)  
“Spectroscopic Binary Detection as Agent-Callable Tools: Detecting 40,000+ Main-Sequence Binary Candidates from SDSS DR19 APOGEE Spectra”  
*The Open Journal of Astrophysics, submitted*
5. **Serat M. Saad**, D. M. Rowan, K. Z. Stanek, et al. (2026)  
“PANTERA. I. Hunting for the Most Metal-Rich Stars in the Solar Neighborhood with High-Resolution Spectroscopy”  
*The Open Journal of Astrophysics, submitted*
4. **Serat M. Saad** & Yuan-Sen Ting (2026)  
“No Gravitational Anomaly in Wide Binaries from Forward Modeling of 3D Orbits”  
*The Open Journal of Astrophysics, 9*
3. **Serat M. Saad** & Yuan-Sen Ting (2026)  
“High-Precision Differential Radial Velocities of C3PO Wide Binaries: A Test of Modified Newtonian Dynamics (MOND)”  
*The Open Journal of Astrophysics, submitted*
2. **Serat M. Saad**, Marina Kounkel, Keivan G. Stassun, et al. (2025)  
“ABYSS III: Observing Accretion Activity in Young Stars through Empirical Veiling Measurements”  
*The Astronomical Journal, 170, 87*
1. **Serat M. Saad**, Kaitlyn Lane, Marina Kounkel, Keivan G. Stassun, et al. (2024)  
“ABYSS II: Identification of Young Stars in Optical SDSS Spectra and Their Properties”  
*The Astronomical Journal, 167, 125*

## CO-AUTHOR PUBLICATIONS

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5. Qinghui Sun, **Serat M. Saad**, Yuan-Sen Ting, et al. (2026)  
“Planets Around Solar Twins/Analogues (PASTA) III: Chemical Clock Relations in Planet-Hosting Solar Analogues”  
*Astronomy & Astrophysics, submitted*
4. SDSS Collaboration, including **Serat M. Saad** (2026)  
“The Twentieth Data Release of the Sloan Digital Sky Surveys”  
*The Astrophysical Journal Supplement Series, submitted*
3. Zimo Cheng, Sharon X. Wang, Fan Liu, ... **Serat M. Saad**, et al. (2026)  
“A Bayesian Search for Planet Engulfment Signatures in Solar Analogues”  
*AAS Journals, submitted*
2. SDSS Collaboration, including **Serat M. Saad** (2026)  
“The Nineteenth Data Release of the Sloan Digital Sky Survey”  
*The Astrophysical Journal Supplement Series, 285, 9*
1. Yuan-Sen Ting, **Serat M. Saad**, Fan Liu, Yuting Shen (2026)  
“Egent: An Autonomous Agent for Equivalent Width Measurement”

## TELESCOPE TIME ALLOCATIONS

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### **Principal Investigator**

|                                                                              |       |
|------------------------------------------------------------------------------|-------|
| Large Binocular Telescope, PEPsi, 20 hours · PANTERA: ultra metal-rich stars | 2026B |
| Large Binocular Telescope, PEPsi, 20 hours · PASTA: 36 planet-hosting stars  | 2026A |
| Large Binocular Telescope, PEPsi, 28 hours · PASTA: 43 planet-hosting stars  | 2025B |

### **Co-Investigator**

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|------------------------------------------------------------------------------|-------|
| Automated Planet Finder, Levy, 40 hours · metal-rich stars (PI: D. M. Rowan) | 2026B |
| Chandra X-ray Observatory, Cycle 26 · Pink Mergers (PI: Prof. Adi Foord)     | 2024  |

## INVITED TALKS AND CONFERENCE PRESENTATIONS

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|                                                                                              |      |
|----------------------------------------------------------------------------------------------|------|
| LLM Agents Seminar, remote                                                                   | 2026 |
| <i>Hands-on session II: agentic coding and research tools</i>                                |      |
| Monday Science Seminar, University of Wisconsin–Madison                                      | 2026 |
| <i>Title to be announced</i>                                                                 |      |
| AAS 245 Winter Meeting, American Astronomical Society                                        | 2025 |
| <i>An Alternative to the Supermassive Black Hole Binary Hypothesis for J1430 (Tick Tock)</i> |      |
| Cool Stars 22, Cambridge                                                                     | 2024 |
| <i>Using Low Resolution Optical Spectra to Identify Young Stars and Their Properties</i>     |      |
| IEEE Quantum Computing and Engineering (QCE)                                                 | 2023 |
| <i>Developing Workforce in Quantum Industry: The Wond’ry Quantum Studio</i>                  |      |

## TEACHING & MENTORING

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|-------------------------------------------------------------------------------------|-------------|
| Mentor, <a href="#">Project SHORT</a>                                               | 2025        |
| Mentor, <a href="#">Bangladesh Olympiad on Astronomy and Astrophysics</a>           | 2021-2024   |
| TA, DS 2100: Statistics for Data Science, Vanderbilt University                     | Spring 2025 |
| TA, ASTR 1010: Introductory Astronomy, Vanderbilt University                        | Spring 2024 |
| Grader, DS 1100: Introduction to Data Science, Vanderbilt University                | Fall 2023   |
| Instructor, Quantum Computing for Everyone, <a href="#">The Wond’ry</a> Vanderbilt  | Fall 2023   |
| Grader, CS 1100: Introduction to Problem Solving with Python, Vanderbilt University | 2022-2023   |
| Tutor, Vanderbilt Student Athletics                                                 | 2022        |

## OTHER EXPERIENCES

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|-----------------------------------------------------------------------------------------|--------------|
| Co-founder, <a href="#">porAI</a>                                                       | 2026-Present |
| Author and Editor, <a href="#">Astrobites</a>                                           | 2026-Present |
| Academic Team Member, <a href="#">Bangladesh Olympiad on Astronomy and Astrophysics</a> | 2021-2024    |
| Co-lead, <a href="#">The Wond’ry</a> Quantum Studio                                     | 2022-2023    |
| Team Leader, <a href="#">International Olympiad on Astronomy and Astrophysics</a>       | 2022         |

## NON-REFEREED PUBLICATIONS

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6. **Serat M. Saad** (2026)  
“‘Total Eclipse of the Heart’ in Astronomy”  
*Astrobites, August*
5. **Serat M. Saad** (2026)  
“Where Have All the Star Couples Gone?”  
*Astrobites, August*
4. **Serat M. Saad** (2026)  
“Cosmic Cannibalism: When Stars Eat Their Planets”  
*Astrobites, May*
3. **Serat M. Saad** (2026)  
“Wormholes Might Be More Real Than We Thought”  
*Astrobites, April*
2. **Serat M. Saad** (2026)  
“A Wandering Supermassive Black Hole Eating a Star”  
*Astrobites, March*
1. **Serat M. Saad**  
“Pixels and Persuasion”  
*As Seen on TV, digital exhibition, Vanderbilt University Libraries*

## REFERENCES

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|                                                    |                                                                                                         |
|----------------------------------------------------|---------------------------------------------------------------------------------------------------------|
| Krzysztof (Kris) Stanek, The Ohio State University | Ph.D. Advisor, <a href="mailto:stanek.32@osu.edu">stanek.32@osu.edu</a>                                 |
| Yuan-Sen Ting, The Ohio State University           | Ph.D. Advisor, <a href="mailto:ting.74@osu.edu">ting.74@osu.edu</a>                                     |
| Marina Kounkel, University of North Florida        | Research Advisor, <a href="mailto:marina.kounkel@unf.edu">marina.kounkel@unf.edu</a>                    |
| Jessie C. Runnoe, Vanderbilt University            | Research Advisor, <a href="mailto:jessie.c.runnoe@vanderbilt.edu">jessie.c.runnoe@vanderbilt.edu</a>    |
| Keivan G. Stassun, Vanderbilt University           | Undergraduate Advisor, <a href="mailto:keivan.stassun@vanderbilt.edu">keivan.stassun@vanderbilt.edu</a> |